INVERTIS UNIVERSITY BAREILLY

Established by Govt. of U.P. u/s 2F of UGC Act, 1956 vide U.P. Act 22 of 2010.

BAS-201- Chemistry

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MODULE-I

Chemical bonding
Molecular Orbital Theory and its applications to Homo and Hetero diatomic molecules, Hydrogen bonding and its consequences, Band theory of metals and its applications.
Liquid crystalline state: Classification and its application. **Solid state**
Solid state:Limiting radius ratio (cubic). Braggs equation.l Distinctive allotropes of carbon such as graphite and fullerenes (two dimensional); properties and applications.
Gaseous state: Gas laws: Boyle‟s law, Charles law, Gay lussac law and kinetic theory of gases.
Reaction kinetics: Order and molecularity of reaction, integrated rate equation for zero first and second order. Theories of reaction rate.
Phase rule: Phase rule and its application to one component system (water)
Electrochemistry
Electrode potential, electrochemical and concentration cell, electrochemical theory of corrosion and itsPrevention.

MODULE-II

Concepts of organics
Electronic displacement in covalent bonded compound, Stability of reaction intermediates; carbocation,carbonation, free radical.E-Z nomenclature and R-S configuration, Conformation of n butane, Nucleophilic substitution reactions Structural and mechanistics
Reaction mechanism of
(i) Aldol Condensation
(ii) CannizarroReaction
(iii) HoffmannRearrangement
(iv) Beckmannrearrangement
(v) Diels Alder reaction.
Polymers
Polymerization techniques; addition, condensation and co – ordination polymerization. Structure preparation, properties and application of Elastomers, plastomers, polyamides and polyesters Conducting Polymers

MODULE-III

Spectroscopy
Elementary idea and simple application of U.V, IR and NMR spectral techniques.
Water: Water processing: boiler feed water (Calgon process), process water (Zeolite process) potable water, (ion exchange method)
Fuel: Analysis of coal (proximate and Ultimate) and their implications, calorific value and its determination (Bomb Calorimeter).

Titrimetric analysis: Types of titrimetric analysis: Acid Base, Redox, Precipitation and Complexometric titrations

Text Books:-

1. *Cotton F.A., Wilkinson G.,Murillo,C.A. and Bochmann" Advanced inorganic chemistry", Wiley,chiester,1992*
2. *Smith,Michael B./March Jerry, March,s "Advanced organic chemistry Reaction , mechanism and structure".,Wilelly and Sons,2007*
3. *Glaston,Samuel B., "Elements of physical chemistry",ELBS,2005Finar,I.L, "Organic Chemistry(vol I&II)" , Addision-Wesley LongmanLtd.*

Reference Book:-

1. *F.W. Billmeyer, "Text Book of Polymer Science", Jonhon Wielly & sons*
2. *G.W. Gray and P.A. Winsor, Ellis "Harwood series in Physical Chemistry, Liquid crystals and plastic crystals (vol I"), ,New York*
3. *M.G. Fontana, "Corrosion Engineering", Mc.Graw Hill Publications.*